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Road Sign Detection and Classification Comparing YOLO and ResNet Architectures

by

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Abstract: This study compares two leading deep learning models for road sign recognition: You Only Look Once (YOLO) for object detection and Residual Neural Network (ResNet) for image classification. A custom dataset comprising four classes—traffic light, stop, speed limit, and crosswalk—was curated and annotated. YOLO was trained to detect and classify road signs in real time using bounding boxes, while ResNet was independently trained on cropped and labeled road sign images.[1] Evaluation metrics included mean Average Precision (mAP), Intersection over Union (IoU), and classification accuracy. The results provide insight into the trade-offs between YOLO's real-time detection capabilities and ResNet's classification precision, offering guidance for future model deployment in intelligent transportation systems.

[1] https://github.com/AliHu264/Road-Sign-Detection/blob/main/YOLO.ipynb

1 Introduction

Road signs are essential to modern traffic systems, informing both human drivers and automated systems. Accurate detection and classification of signs is critical for safety and regulatory compliance. This project evaluates two deep learning models, YOLO and ResNet, applied separately to road sign recognition tasks.

- Implementing YOLO for real-time detection and classification.
- Training a ResNet classifier on labeled cropped road sign images.
- Comparing model performance across various metrics.

2 Related Work

Traditional methods for road sign recognition used features like HOG, SIFT, and SVMs. With deep learning, architectures like Faster R-CNN, SSD, and YOLO significantly advanced detection capabilities. ResNet introduced deep residual learning, enabling the training of very deep models. This work contrasts YOLO's detection-centric design against ResNet's image classification approach.

3 Dataset

The dataset consists of annotated images from four classes: traffic light, stop, speed limit, and crosswalk. Data were split into training, validation, and test sets. YOLO annotations used the .yaml configuration format, while ResNet used labeled cropped images.

4 Methodology

4.1 YOLO Detection

YOLO was trained using the annotated dataset to detect bounding boxes and classify signs simultaneously. Training was optimized for mAP and IoU with SGD.

4.2 ResNet Classification

ResNet was trained separately on manually cropped sign images. The network was fine-tuned using cross-entropy loss and an SGD optimizer.

5 Experiments

5.1 Training Setup

- **YOLOv8n:** Multiple runs with pretrained and scratch weights.
- **ResNet-50:** Fine-tuned for classification on cropped signs.

5.2 Evaluation Metrics

- **YOLO:** mAP@0.5, precision.
- **ResNet:** mAP@0.5, confusion matrix.

5.3 ResNet Training

Images were resized to 224×224, normalized, and augmented (flipping, brightness, rotation). The model was trained with a batch size of 32 over 10 epochs using SGD optimizer ($lr = 0.0001$).

5.4 Results

- **YOLO - Pretrained, no fine-tuning:** mAP@0.5 = 0.00077
- **YOLO - Pretrained, 5 epochs:** mAP@0.5 = 0.741
- **YOLO - Scratch, 5 epochs:** mAP@0.5 = 0.151
- **YOLO - Scratch, 15 epochs:** mAP@0.5 = 0.494
- **ResNet-50 (fine-tuned, 5 epochs):** mAP@0.5 = 0.855

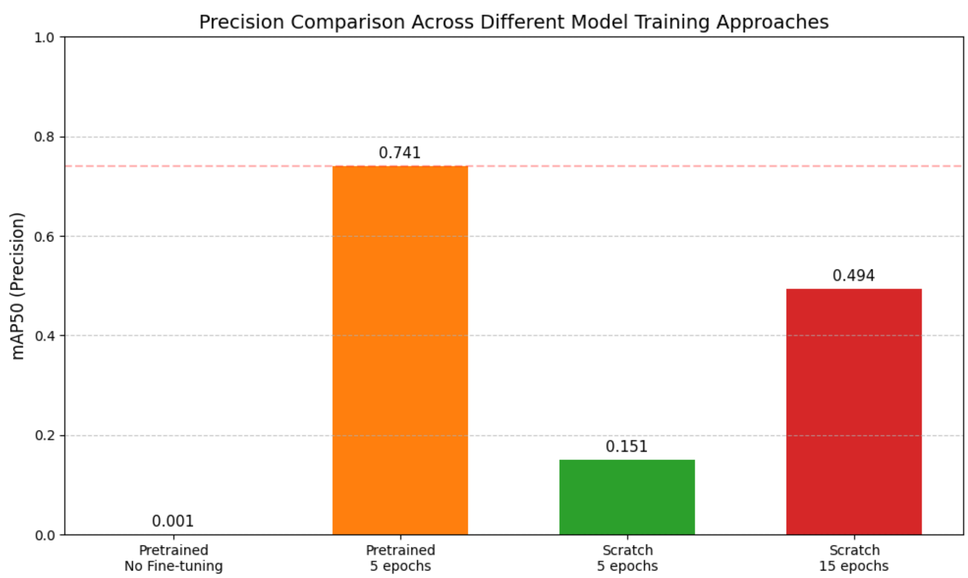

Figure 1: YOLO precision comparison across training configurations.

Figure 2: YOLO detection output on test images.

Figure 3: ResNet classification on test images.

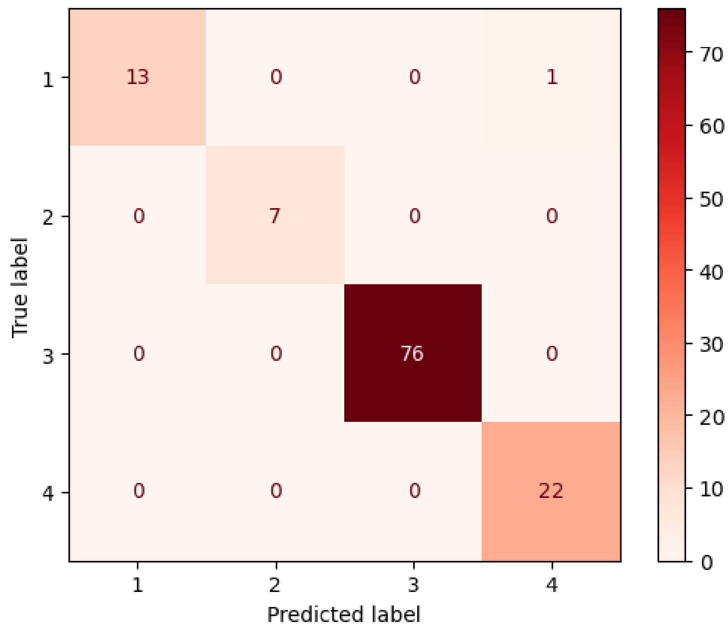

Figure 4: ResNet confusion matrix on test set.

6 Conclusion

This project compared YOLOv8n and ResNet-50 on a custom road sign dataset. YOLO excelled in real-time detection and low computational demand, while ResNet achieved higher classification accuracy. These models serve different operational purposes. YOLO is suited for detection-heavy real-time environments; ResNet is preferable when classification precision is critical.

Another item of note is the speed of performance in YOLO is superior to that of the ResNet model.

A key limitation observed was the models' behavior when exposed to unfamiliar inputs. In one case, YOLO detected and labeled a road sign that was not part of the four defined classes (traffic light, stop, speed limit, crosswalk), resulting in a false positive. This highlights a critical challenge in model generalization—when data outside the trained distribution is encountered, incorrect detections may occur. To address this, future iterations should incorporate a more extensive and diverse dataset encompassing a broader range of road sign types.

Future work may explore integrating both models into a unified pipeline, but this study restricts itself to isolated performance comparison.